

I Sets & Functions Lecture 6 II Writing III Divisibility IV Modular Arith

Cantor Proof: For every sequence $x_1, x_2, x_3, \dots, x_n \in (0,1)$
 There exists an $x \in (0,1)$
 s.t. $x \neq x_n$ for all n .

<demo>.

□

"There are different sizes of infinity."

$$|\mathbb{Z}| < |(0,1)|$$

Summary: • A, B countable $\rightarrow A \cup B$ countable

• $A \subseteq B, B$ countable $\rightarrow A$ countable

• A, B countable $\rightarrow A \times B$ countable

$a_1 a_2 \dots$
 $b_1 b_2 \dots$
 $\downarrow$
 $a_1 b_1, a_2 b_2, \dots$
delete elements
in $B-A$.
list diagonals.

HW: "There are infinitely many infinities." $\forall A$ $|A| < |\mathcal{P}(A)|$.

Loose Ends:

Prop: If $f: A \rightarrow B$ and $f^{-1}: B \rightarrow A$ then

$$\left. \begin{aligned} (f^{-1} \circ f)(a) &= a \quad \forall a \in A \\ (f \circ f^{-1})(b) &= b \quad \forall b \in B. \end{aligned} \right\}$$

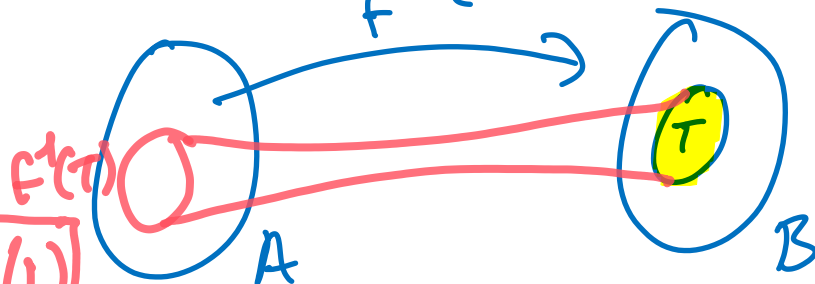
ex: $f: \mathbb{R} \rightarrow \mathbb{R}_{>0}$ $f(x) = e^x$
 $f^{-1}: \mathbb{R}_{>0} \rightarrow \mathbb{R}$ $f^{-1}(y) = \ln(y)$
 $\ln(e^x) = x$; $e^{\ln y} = y$.

Defn: If $f: A \rightarrow B$ and $T \subseteq B$ then

$$f^{-1}(\underline{T}) = \{ a \in A : f(a) \in T \}.$$

Not the
same as
inverse!

cf. $f^{-1}(b)$, $b \in B$.



$f^{-1}(T)$ different from $f^{-1}(b)$

II Writing

- A proof has a

Beginning { Assume hypotheses.
Assume x is arbitrary.
Middle { Observe that ...
Consider y ...
End { We conclude that...
...As desired.

- Use keywords to indicate logical structure
(Assume, ...arbitrary, choose, by defn.)

Proof: (linear) Assumptions $\xrightarrow{\text{small, easily checkable}} \dots \rightarrow$ Conclusion.
Defns

Process: (messy) Assumptions $\xrightarrow{\text{check}} \dots \xrightarrow{\text{check}} \dots \rightarrow$ Conclusion
Defns

"Distillation of thought process."

Scratchwork
plan
draft
edit
rewrite.

How to solve "Prove or Disprove S".

Step 1: Write down S and $\neg S$.

Step 2: Write down all defs

Step 3: Form a belief about S or $\neg S$.

↙
examples

↓
similar
statements.

↘
counterexamples.

Step 4: Try to write a proof.

↙ success
😊

→ fail.

What did you learn about S ?
Change proof strategy?
Counterexample?

Repeat.

Ways to use AI:

- ① Explain a defn, proof. — NotebookLM.
Hazard: Might be wrong.
Must be able to read + check proofs.
- ② Check your work. — GPT-4o — gives useful feedback.
Hazard: Can be wrong.
Don't learn how to check.
- ③ Stuck. — Not learning how to find proofs.

So far:

Logic + Proofs (3).

Sets & Functions (2).

Number Theory (4) : Study of $\mathbb{Z}$.

I Division.

Defn: If $a \neq 0$ and b are integers, a divides b denoted $a|b$ if there is an integer k such that $b = ka$.
 $a|b \leftrightarrow \exists k(b = ka)$.

ex: $3|9$ $3|0$ $3 \nmid 10$ $3|-9$
 $9 = 3 \cdot \underset{k}{3}$, $0 = 0 \cdot \underset{k}{3}$, $\forall k(10 \neq 3k)$, $-9 = 3 \cdot \underset{k}{(-3)}$.

(Q) $0|b \leftrightarrow \exists k \ b = 0 \cdot k$. Not a useful defn.

Prop 1: $\forall a, b, c \in \mathbb{Z}$

- ① if $a|b$ and $b|c$ then $a|c$.
- ② if $a|b$ and $a|c$ then $a|b+c$.

Proof:

- ① <HW>
- ② Assume $a, b, c \in \mathbb{Z}$ are arbitrary.
Assume $a|b$ and $a|c$.

a, b, c
 $a|b, a|c$
 k_1, k_2

By defn, we can choose $k_1, k_2 \in \mathbb{Z}$ s.t.
 $b = ak_1$ and $c = ak_2$.

Adding these equations,

$$b + c = ak_1 + ak_2 = a(k_1 + k_2) \in \mathbb{Z}$$

By defn, $a|b+c$ as desired. $\square$

Q) What is division?

$$\begin{array}{r} \text{quotient} \\ 3 \overline{) 20265} \\ \vdots \end{array}$$

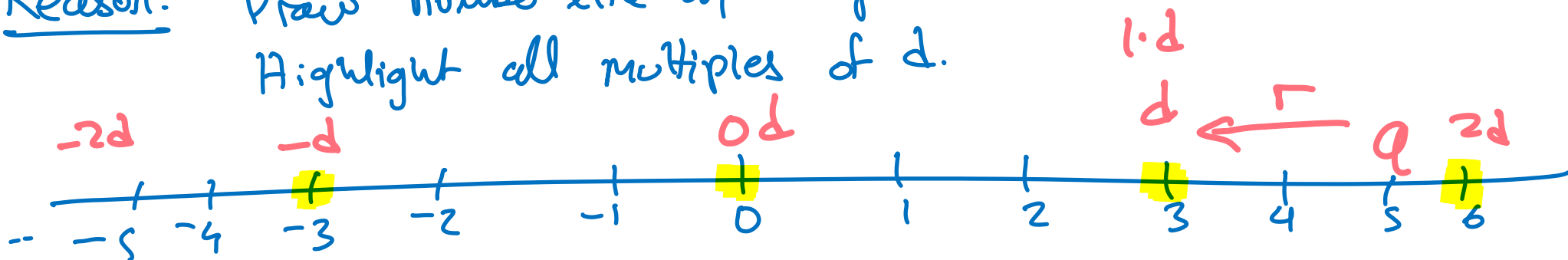
remainder.

Theorem: (Division Algorithm) If $a \in \mathbb{Z}$ and $d \in \mathbb{Z}^+$ then
 there exist unique $q, r \in \mathbb{Z}$ such that
 q ~~quotient~~ ~~remainder~~

① $a = dq + r$

② $0 \leq r < d$.

Reason: Draw number line w/ integers:
Highlight all multiples of d .



There is a first highlighted number
which is to the left of a (could be a itself)

That number is dq for some $q \in \mathbb{Z}$.

Observe:

$$a = dq + r$$

$$0 \leq r < d.$$

(Q) Is $\text{division}(a, d) = (q, r)$ onto? <later>

To prove formally, need:



Axiom : [Well-Ordering Principle]: Every nonempty subset $S \subseteq \mathbb{Z}_+$ has a least element.

$$\forall S (S \subseteq \mathbb{Z}_+ \wedge S \neq \emptyset \rightarrow \exists y \in S (\forall x \in S (x \geq y)))$$

Theorem: (Division Algorithm) If $a \in \mathbb{Z}$ and $d \in \mathbb{Z}_+$ then there exist unique ^{quotient} $q, r \in \mathbb{Z}$ ^{remainder} such that

- ① $a = dq + r$
- ② $0 \leq r < d$.

Proof of Thm: Assume $a \in \mathbb{Z}$ and $d \in \mathbb{Z}_+$ are arbitrary.

Consider $S = \{ a - dq : q \in \mathbb{Z} \text{ and } a - dq \geq 0 \} \subseteq \mathbb{Z}_+$
(encodes multiples of q to the left of a).

Observe that $S \neq \emptyset$:

mini proof $\left\{ \begin{array}{ll} \text{if } a \geq 0 & \text{then letting } q=0: a-dq=a \geq 0. \\ \text{if } a < 0 & \text{then letting } q=a: a-d \cdot a = (1-d)a \geq 0. \end{array} \right.$

By the WOP: we can choose a least element $r \in S$.
 ("First multiple of q to the left")

By construction, there exists $q \in \mathbb{Z}$ such that $a-dq=r$,
 showing ①.

Observe that $r \geq 0$ by the defn of S .

Assume for contradiction that $r \geq d$.

mini proof $\left\{ \begin{array}{l} \text{Then } 0 \leq r-d = a-dq-d = a-d(q+1) \\ \text{so } r-d \in S, \text{ contradicting the minimality of } r. \end{array} \right.$

This shows existence.

Uniqueness < HW?.

□

Notation: If $a = dq + r$, $0 \leq r < d$ then

$$\begin{aligned} a \text{ div } d &= q && \text{quotient} \\ a \bmod d &= r. && \text{remainder.} \end{aligned}$$

II Modular arithmetic.

Ex 1: I sleep at 11pm for 8 hrs, I get up at ~~11pm~~ 7am.
 $19 \equiv 7 \pmod{12}$

Defn: If $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}_+$ then a is congruent to b modulo m denoted $a \equiv b \pmod{m}$ if $a \bmod m = b \bmod m$.

"notion of equivalence which only cares about the remainder mod m ."

Prop 2: $\forall a, b \in \mathbb{Z} \ m \in \mathbb{Z}_+, \quad a \equiv b \pmod{m} \iff m \mid b-a.$

Proof: ($\rightarrow$) Assume $a \pmod{m} = b \pmod{m}.$

We can choose q_1, q_2, r s.t. $q_1 : a = mq_1 + r$
 $q_2 : b = mq_2 + r.$

Subtracting, $b-a = m(q_2 - q_1) + \cancel{(r-r)},$
so $m \mid b-a.$

($\leftarrow$) Hw.

□

Prop 3 (Properties of $\equiv$)

① If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$
then $a \equiv c \pmod{m}$.

(substitution).

② If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$
then

① $ac \equiv bd \pmod{m}$

② $a+c \equiv b+d \pmod{m}$.

(can add/multiply
 $\equiv$ -equations).

Do you use AI for learning math? If so, how?

